

SAFETY QUIZ: CHE381- Chemical Engineering Laboratory I

Please answer the following questions testing your knowledge of the content of the Chemical Hygiene Plan safety document. Be brief but concise in your answers. Open book/open notes and team discussion is allowed but answers must be in your own words.

1. What is the most fundamental rule of chemical safety?
2. What two pieces of personal protective equipment must be worn at all times in the lab?
3. What additional kinds of personal protective equipment are available depending on the chemical hazard or experimental conditions?
4. True or False: Fume hoods are generally used as storage areas for chemicals.
5. Fume hoods are to be used when the Threshold Limit Value (TLV) of a chemical is at or below:
6. What are the four routes of chemical entry into the body?
7. True or False: Respirators are generally used in the R&D laboratories for normal work procedures.

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8. What is the role of the Chemical Hygiene Plan?

9. What is the guideline for laboratory room airflow and fume hood airflow?

10. List 3 Common Housekeeping Practices that promote chemical hygiene and safety:

11. What is the proper method of transporting gas cylinders in the laboratory?

12. True or False: MSDS sheets are required only for toxic chemicals and gases used in the laboratory.

13. What is an accepted method of vapor detection in the laboratory?

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14. Name 3 Personal Hygiene Practices that should be followed in the laboratory.
15. Describe how individual chemicals of less than 15 lbs. (for example a bottle of acetone) are safely transported.
16. List the Emergency Procedure to be followed for a large chemical spill or toxic gas release in the laboratory.
17. What is the difference between a flammable chemical and a combustible chemical in terms of OSHA and NFPA guidelines.

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18. List the Emergency Procedure for an individual chemical exposure in the laboratory.

19. What is the definition of a “hazardous chemical”?

20. All laboratories that handle chemicals must be equipped with eyewashes and safety showers. What is the recommended flowrate of this equipment and how often should they be checked?

BONUS QUESTION:

Where are the fire extinguishers located in the Unit Operations Laboratory (Room 120)?